

A total of 76 engineering students and professionals from all over India participated in the event. The 'breeding hackers' team had Vivek Khurana (Delhi, the guest of honour), Atul Jha (Delhi, *mukt.in*), Gajendra Khanna (student coordinator, IIT-Delhi LUG), Mayank Kumar (student coordinator, IIT-Delhi LUG), Ajay Kumar (Sahana, GSOC '08), Pranava Swaroop (NIT-Jaipur), P. Govind Raj and Jai Bhaskar (both from the BOSS team of CDAC-Noida).

Media partners Radio IIMT-90.4 FM, the first community radio of North India, showed true community spirit by recording the historic promo for FOSSTIVAL 2009. The promo clearly defined a 'hacker' as: "A hacker is not a criminal, but instead, an artistic programmer! Come and become a true hacker at the IIMT FOSSTIVAL workshop." Hats off to the director of 90.4 FM, who took a bold step towards spreading the truth about the 'hacker culture' among the public. The promo has been released under the Creative Commons Share Alike License on www.glug-meerut.org. The director of 90.4 FM, Braham Prakash Yadav, is working on creating an English version

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of the promo to be released under the Creative Commons. Richard Stallman wrote the following lines for FOSSTIVAL Promo at Radio IIMT 90.4 FM: "Hacking doesn't mean breaking security, Hacking means playful cleverness. Come join us in hacking!"

Day 1: Booting it up

The first day (January 9) started with a bang! It was an unusual inauguration of a true hacker's event—bursting more than a dozen balloons with names of different non-free software on them! The first balloon was burst by guest of honour Vivek Khurana. This '*Halla Bol*' inauguration thrilled the budding GNU/Linux hackers!

Vivek Khurana took charge of the Web development track by recapitulating the science and technology behind the magic of the Web. He covered the HTTP client and server system, and how they run in a global scenario, the mark-up languages and their role in the Web, W3C, IETF and showed the role of hacking scripting languages for the Web ecosystem. The emphasis was on understanding the protocol standards and development standards for the Web.

Atul Jha initiated the GNU/Linux System Programming track with an overview of the system, shell tips and tricks, as well as a little dose of BASHing. The ground was set for a closer look at the software development ecosystem of the GNU/Linux system using a variety of software tools.

Day 2: Getting the kernel going

The Web track was started by Pranava Swaroop, who demonstrated how to 'configure, learn and run' the grand Apache server. Starting with the internals of Apache and the design of the Apache Web server, Pranava shared some tricks on installing, configuring, hosting and even securing the Apache server. The Apache kernel, *http_core* and libraries excited the young hackers. Other topics of interest with the audience were virtual hosts and *mod_ssl* usage.

Vivek Khurana followed, with an introduction to dynamic languages and the PHP execution model. He then moved on to a hands-on session on PHP. Participants developed PHP applications using the MVC paradigm. Vivek also shared the nuts and bolts of PHP like the data types, sessions, cookies, and OOP. The session ended with an introduction to MySQL and Web services.

In the systems programming track, Gajendra Khanna (IIT Delhi LUG) used the ground set by Atul to interact with GNU/Linux software tools. He started with the famous 'vi versus Emacs' and could not resist interacting with his favourite 'Emacs Psychiatrist' while comparing the two. The Eclipse IDE workhorse was a bonus for the audience. Gajendra demonstrated the intricacies of the GNU Toolchain members using simple programs. He covered GCC, Make, GDB and CVS, which seldom feature in workshops as harmonious elements of a standard toolchain.

Mayank Kumar (IIT Delhi LUG) followed Gajendra by initiating practical IEEE POSIX sessions. He moved from the Linux system call interface and went on to the file system interface, processes and POSIX threads, as well as covering a bit of signals and their usage. The hands-on creation and usage of processes and *Pthreads* gave enough food for thought to budding GNU/Linux systems programmers. The issue of portable applications was discussed in the light of POSIX. Several participants became eager to contribute to the systems programming-oriented open source projects.

Evening 2: Google Summer of Code in the winter of Meerut

In a special evening session, Ajay Kumar (Google Summer of Code winner of 2008) introduced project Sahana—the disaster management system, as a group member of the Sahana core development team. He discussed the brief history of Sahana—the humanitarian FOSS project, the problems faced by it, the technologies used, the deployment scenario, real-life deployments and case studies.

Ajay discussed the Sahana Request Management System, Messaging System and Real-Time Situation Maps with an emphasis on deployment aspects. A special attraction was 'Sahana on Mobiles'.

He inspired young hackers to take up the challenge and opportunities presented by the 'Google Summer of Code'. He discussed the goals of the program, how it benefits the student community at large, what students could do with it and how it benefits them in terms of student stipends.